Genomics 112 (2020) 1371–1383

Contents lists available at ScienceDirect

Genomics

journal homepage: www.elsevier.com/locate/ygeno

Original Article

Genome-wide identification and expression analysis of two component system genes in *Cicer arietinum*

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ARTICLE INFO

Keywords:
TCS genes
Phylogenetics
Evolutionary analysis
Abiotic stress
Chickpea

ABSTRACT

The two-component system (TCS) plays an important role in signal transduction pathways, cytokinin signaling and stress resistance of prokaryotes and eukaryotes. It is comprised of three types of proteins in plants; histidine kinases (HKs), histidine phosphotransfer proteins (HPs) and response regulators (RRs). Chickpea (*Cicer arietinum* L.) is one of the most important legume crops worldwide with special economic value in semi-arid tropics. Availability of complete genome sequence of chickpea presents a valuable resource for comparative analysis among angiosperms. In current study, *Arabidopsis thaliana* and *Oryza sativa* were used as reference plant species for comparative genomics analysis with *C. arietinum*. A genome-wide computational survey enabled us to identify putative members of TCS protein family including 18HKs, 26 RRs (7 type-A, 7 type-B, 2 type C and 10 pseudo) and 7 HPs (5 true and 2pseudo) genes in chickpea. The predicted TCS genes displayed family specific intron/exon organization and were randomly distributed across all the eight chromosomes. Comparative phylogenetic and evolutionary analysis suggested a variable conservation of TCS genes in relation to mono/dicot model plants and segmental duplication was the principal route of expansion for this family in chickpea. The promoter regions of TCS genes exhibited several abiotic stress-related *cis*-elements indicating their involvement in abiotic stress response. The expression analysis of TCS genes demonstrated stress (drought, heat, osmotic and salt) specific differential expression. Current study provides insight into TCS genes in *C. arietinum,* which will be helpful for further functional analysis of these genes in response to different abiotic stresses.

1. Introduction

Two-component system (TCS) was initially discovered in bacteria, but it has been shown to play an important role in different signal transduction pathways in slime molds, fungi and plants [1–4]. In prokaryotes, this system typically involves a Histidine Kinase (HK) (a membrane-localized protein receptor) and a Response Regulator (RR) [4,5]. When HK receives an environmental stimulus, it autophosphorylates the histidine (H) residue in the kinase acceptor domain (Hiska). Afterwards, the phosphoryl group is transferred to the aspartic acid (D) residue of the receiver (Rec) domain of RRs proteins [5]. With the passage of time, a multistep phosphorylation system has been evolved in eukaryotes, which involves an additional protein group known as histidine phosphotransfer proteins (HPs) [6]. The members of HP family transfer the phosphate group and act as a bridge between HK and RR proteins in the signaling cascade [4,7,8]. There are three subfamilies of HKs in plants as cytokinin receptors, ethylene receptors and phytochromes [9]. Moreover, there are three additional HKs (AHK1,

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https://doi.org/10.1016/j.ygeno.2019.08.006
Received 11 June 2019; Received in revised form 19 July 2019; Accepted 11 August 2019
Available online 12 August 2019
0888-7543/

ACKI1 and AHK5/CKI2), which are not part of any group in *Arabidopsis* [9]. The overall structure of HKs contains an input domain, several trans-membrane domains (present at N-terminus), a transmitter domain containing a conserved H residue (which is the site of auto-phosphorylation) and a receiver (Rec) domain [9]. However, transmitters of three ethylene receptors (AEIN4, AETR2 and ERS2) and phytochrome are unlikely to perform the HK activity because they lack highly conserved residues and motifs. Therefore, they are known as divergent HKs [10–13]. Furthermore, AHK2, AHK3 and AHK4, which are cytokinin receptors, contain a cyclase/HK-associated sensory extracellular (CHASE) domain that is recognized by cytokinin 14. ERS1, ERS2, AER1, AETR2 and AEIN4 are the ethylene receptors which containan ethylene (C_2H_4) binding domain. In addition, members of the phytochrome family (PHYA, PHYB, PHYC, PHYD and PHYE) contain two PAS (Per/Arndt/Sim) folds and a chromophore-binding (PHY) domain [4].

Phosphotransfer (Hpt) domain of HP family contains a highly conserved motif (XHQXKGSSXS) necessary for transfer of phosphate group from Rec domain of HKs to the Rec domain of RRs [9]. However, conserved H residue is missing in AHP6. Therefore, it is known as a pseudo-His-containing phosphotransfer protein (Pseudo-Hp). Moreover, AHP6 is a negative regulator of cytokinin signaling and cannot act as a phosphotransfer protein [10]. Based on conserved sequences, signatures and domains, there are three subgroups of RR family: type-A, type-B and type C. Primarily, type-A RRs are cytokinin response proteins and contain C-terminal extensions along with the Rec domain. On the other hand, type-B RRs are transcriptional factors consisting of a C-terminal output domain and an N-terminal Rec domain [11]. Additionally, type-C RRs also have same domain structure as type-A RRs, but they are not induced by cytokinin [12] and to the date no specific role of type-C RR is known in cytokinin signaling [12–14]. Furthermore, a divergent class of RRs is present and these RRs are known as pseudo-RRs (PRRs). A highly conserved D residue, necessary for phosphrylation, is missing in PRRs [15]. PRRs contain a specific CCT (Co, Col and Toc1) motif in their C-terminal extension, which plays a vital role in the regulation of circadian rhythms [16–18].

The TCS have been involved in numerous processes related to growth, development and adaptation like nutrient sensing, chemotaxis, hormone signaling, stress response, nodulation and endosperm formation [4,19–22]. TCS genes have been extensively studied for signaling related to cytokinin, ethylene and phytochromes [6,23–26]. Cytokinin (CK) hormone plays an important role in different aspects of plant life [25,27–29]. Among different classes of CKs, isopentenyladenine-type and trans-zeatin CKs, have been studied for functional properties. The role of cis-zeatin-type CKs (cZs) have been less well defined [30]. Majority of previous studies present the differences of CK profiles between model and crop plants. However, *Arabidopsis* contain lower amounts of cZs than *trans* isomers and its CK receptors possess lower affinity for cZs as compared to other plants [31]. Therefore, we might need novel systems for the functional predictions of cZs type CKs. Interestingly, cZs are abundantly present in rice, maize and chickpea [27,32,33]. Previous studies have suggested a distinct role of CKs and a differential ligand preference for CK receptors in maize [31]. Yet, scanty information is available for molecular identities of CK signaling in important crops like chickpea. TCS genes are involved in stress response as positive regulators (*AHK1, OsHK3, GmHK7, GmHP3, GmHP6, GmRR1*) or negative regulators (*AHK2, AHK3, AHK4, AHP1, AHP3, AHP5, ARR8, and ARR9, OsHK4, GmHK10, GmHK12, GmPHP2*) during drought stress [34–37]. In chickpea, the only reported TCS gene is *CarETR1* (*Cicer arietinum* L. ethylene receptor-like sequences), which is linked with chickpea resistance to *Ascochyta rabiei* [38]. Later, the *CarETR1* alleles were used for chickpea breeding to eliminate susceptible individuals [39].

C. arietinum is one of the most important legume crops and it contains high contents of carbohydrates and proteins. High nutritional and economical value makes it extremely important for food security, especially in developing countries. This crop is generally grown in nutrient poor soils under harsh climatic conditions [40]. Through long-term evolution and adaptation to extreme conditions, chickpea has been found to be rich in resistance genes for a range of abiotic stresses, including drought and cold, and has been suggested as a model plant for investigation of physiological mechanisms of plant development and responses to stresses [41,42]. However, there has been no report about the identification and characterization of *TCS* genes from chickpea. Recently, genome sequence of chickpea has been made available in public databases [42–44], which enabled us to retrieve the sequences and predict the functions of TCS gene families via comparative genomics and transcriptomics.

2. Materials and methods

2.1. Identification of TCS genes in C. arietinum

To retrieve the protein and DNA sequences of *TCS* genes in *C. arietinum,* protein blast (BLASTp), PSI-BLAST and DELTA-BLAST were performed against chickpea genome database and Genbank. For this purpose, *Arabidopsis* TCS protein sequences were used as a query. All the variants were checked carefully (*E*-value less than $1e^{-10}$) and the variants containing the longest open reading frames (ORF) were selected for further analysis. Afterwards, the target sequences were processed by using different databases i.e. NCBI conserved domain database (http://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi) [45], SMART database (http://smart.embl-heidelberg.de/) [46] and Pfam database (http://pfam.janelia.org/) [47]. This processing was performed to eliminate the sequences that do not contain conserved domains, which are necessary for normal functionality of TCS proteins. Genomic information related to the *TCS* genes i.e. chromosomal location, length of protein, and their genomic and cDNA sequences, was taken from NCBI.

2.2. Gene structure and prediction of Cis regulatory elements

Gene structure analysis and their schematic representation were done via an online freely available server known as Gene Structure Display Server (http://gsds.cbi.pku.edu.cn/) [48]. For prediction of cis regulatory elements, the upstream sequences (1000–1200 bp) of the *TCS* genes were analyzed. Identification of cis regulatory elements along with the promoter sequences was done via the PlantCARE website (http://bioinformatics.psb.ugent.be/webtools/plantcare/html/search_CARE.html) [49].

2.3. Motif recognition, phylogenetic analysis, and multiple sequence alignment

Phylogenetic analysis was performed by using MEGA 7.0 software [50]. Both Maximum Likelihood and Neighbor-joining (NJ) methods were used for tree construction and other parameters that were taken into consideration were pairwise deletion, poisson correction and bootstrapping (1000 replicates). For phylogenetic analysis, sequences of HKs, HPs and RRs were taken from *Arabidopsis, O. sativa* and *C. arietinum*. Sequences of different conserved domains (Rec, Myb, Hpt and HK) and motifs i.e. CCT were used for multiple sequence alignment via ClustalX program [51] to study the conservation of important residues that are involved in phosphorylation mechanism.

2.4. Chromosomal mapping, gene duplications and evolutionary analysis of the TCS members in C. arietinum

All the TCS members were mapped on the 8 chromosomes of *C. arietinum* by using Mapchart program [52]. Tandem or segmental, the type of duplication was found by using the online tool (http://chibba.agtec.uga.edu/duplication/) [53]. Occurrence of duplication events, selective pressure on duplicated genes and divergence of homologous

Table 1
Number of TCS genes in different plant species.

Species	HK	Hpt	Type-A RR	Type-BRR	Type-C RR	Pseudo-RR	Total	References
Arabidopsis	8	6[a]	10	12	2	9	47	[3]
O. sativa	8	5[b]	13	13	2	8	49	[57]
G. max	21	13	18	15	3	13	83	[58]
L. japonicas	14	7	7	11	1	5[c]	40	[59]
P. patens	18	3	7	5	2	4[c]	39	[60]
C. arietinum	18[f]	7[d]	7	7	2	10[e]	51	Present work

[a] Five authentic and one pseudo-HPts.
[b] Two authentic and three pseudo-HPts.
[c] Only clock-associated.
[d] Five authentic and two pseudo-HPts.
[e] Both clock associated and type-B PRRs.
[f] 10 true HKs and 8 divergent.

genes were estimated by finding the Ka (non-synonymous) and Ks (synonymous) substitution rate per site between the gene pairs by using an offline tool DNAsp. Divergence time was calculated by using (1.5×10^{-8} substitution/synonymous site/year) as a synonymous substitution rate for dicots [54].

2.5. Plant material and stress imposition

The National Institute of Agriculture and Biotechnology (NIAB, Faisalabad, Pakistan) provided the seeds of chickpea (genotype K-70005, Kabuli type). An environmental control chamber was used to grow plants in peat moss under following conditions: 65% relative humidity, 22/20 °C day/night temperature and 8/16 h dark/light period. Stress treatments (drought, heat, osmotic or salinity) were applied when seedlings were at the age of 15 days after germination. To enforce drought stress, limited water was provided to experimental group as compared to an ample water supply for control plants over a period of 10 days. For salt stress treatment, 100 mM NaCl solution was directly applied in pots of experimental group for a period of 72 h. For osmotic stress, 100 mM mannitol solution was added in pots of experimental group for 72 h. For heat stress treatment, plants were kept in environment-controlled chamber at 42 °C for a period of 24 h [55]. RNA was extracted from leaf samples that were previously frozen in liquid nitrogen and were stored at −80 °C.

2.6. Expression profiling of TCS genes in C. arietinum

Expression data of the *TCS* genes was downloaded from the chickpea transcriptome database (CTDB) 55. For Real-time qPCR analysis, RNeasy Plant Mini Kit (QIAGEN: Cat No./ID: 74904) was used for total RNA extraction from leaf samples. NanoDrop spectrophotometer (Colibri spectrometer, Titertek Berthold, Germany) was used for quantification of RNA. Maxima H Minus First Strand cDNA Synthesis Kit, with dsDNase (cat#K1681) was used for cDNA synthesis from one microgram total RNA. iTaq Universal SYBR Green Super Mix was used for qRT-PCR analysis using CFX96 Touch™ Real-Time PCR Detection System. An internal control gene (*CarGAPDH:* XM_004515773.2, F-primer, 5′- GAAGCTTGAGAAGGCCGCTA-3′; R-primer, 5′-TGCCTTTCA ACTTGCCCTCA-3′) was used for normalization of the expression data [56]. The gene-specific primers were designed by using the online tool "Oligo Calculator" (http://mcb.berkeley.edu/labs/krantz/tools/oligocalc.html) and primer specificity was verified by NCBI Primer-BLAST program (https://www.ncbi.nlm.nih.gov/tools/primer-blast/). The qPCR analysis was repeated in three independent experiments using three biological repeats of each treatment. Sample preparation and qRT-PCR analysis were conducted following manufacturer's instructions. A 10 ul of reaction mix contained of SYBR green (1×), forward and reverse primers (10 mmol each) and template cDNA(10 ng) from reverse transcriptase PCR. An equal amount of cDNA template was used for each sample including the internal control. PCR amplification conditions were as follows: 1) initial denaturation (10 min at 95 °C); 2) denaturation (10 s at 95 °C), annealing (20 s at 58 °C), extension (30 s at 72 °C) for 40 cycles; and 3) melting curve analysis to confirm the specificity of the PCR product.

3. Results and discussion

3.1. Identification of TCS genes in C. arietinum

A total of 51 genes were found on the basis of BLASTp search in NCBI. The genomic, protein and CDS sequences are provided in supplementary dataset 1.

3.1.1. HK family

Current genome wide analysis revealed the presence of 18 HKs (CarHKs/CarHKLs) in the genome of chickpea, which is higher than *Arabidopsis* (8) and *O. sativa* (8). However, this higher number of HKs is comparable with other legumes like *G.* max (21) and L. *japonicus* (14), which signifies potential importance of HKs in these plants (Table 1). In chickpea, only 10 HK members (named as true HKs) have the conserved residues required for HK activity and the other 8 members were placed in the divergent HKs group (Table 1 & Table S1). The cytokinin receptor family in *C. arietinum* has four members i.e. CarHK2, CarHK3, CarHKL3 and CarHK4 and contain trans-membrane domains, HisKa domain, CHASE domain (involved in cytokinin binding), and a response regulator Rec domain (Fig. 1). CarHK2 is a divergent HK as conserved domain analysis indicates that all members contain conserved motifs except CarHK2, in which glycine (G) residue in G1 motif is replaced by glutamate (E) (Supp. Fig. 1).

All five motifs are conserved in CarCKI1, which shows its relation to the true HK family (Supp Fig. 1). In *Arabidopsis*, it was thought that CKI1 is a hybrid HK with a conserved Rec and Hiska domain involved in cytokinin signaling [61]. In yeast, failure of cytokinin binding to CKI1 indicates that it is not involved in cytokinin signaling [62]. Domain analysis in current study indicates that CarCKI1 lacks CHASE domain (Fig. 1). Therefore, it was not placed in the cytokinin family (Table S1, Fig. 1).

CarCKI2 contain a conserved HisKa and a C-terminal Rec domain, but lack N-terminal trans-membrane domains. Two members (CarHK5 and CarHKL5) were placed in CKI2 group in *C. arietinum* as they lack N-terminal trans-membrane domains. They are declared as true HKs because most of conserved motifs are present in their protein sequence (Supp Fig. 1). AHK1 in *Arabidopsis* contains two N-terminal trans-membrane domains, a Hiska domain and a C-terminal Rec domain. Two members (CarHK1 and CarHKL1) were identified in *C. arietinum* that contain the same domain structure as their *Arabidopsis* counterpart (AHK1). They are considered as true HKs as they also contain all the conserved residues necessary for normal functioning of HKs (Supp

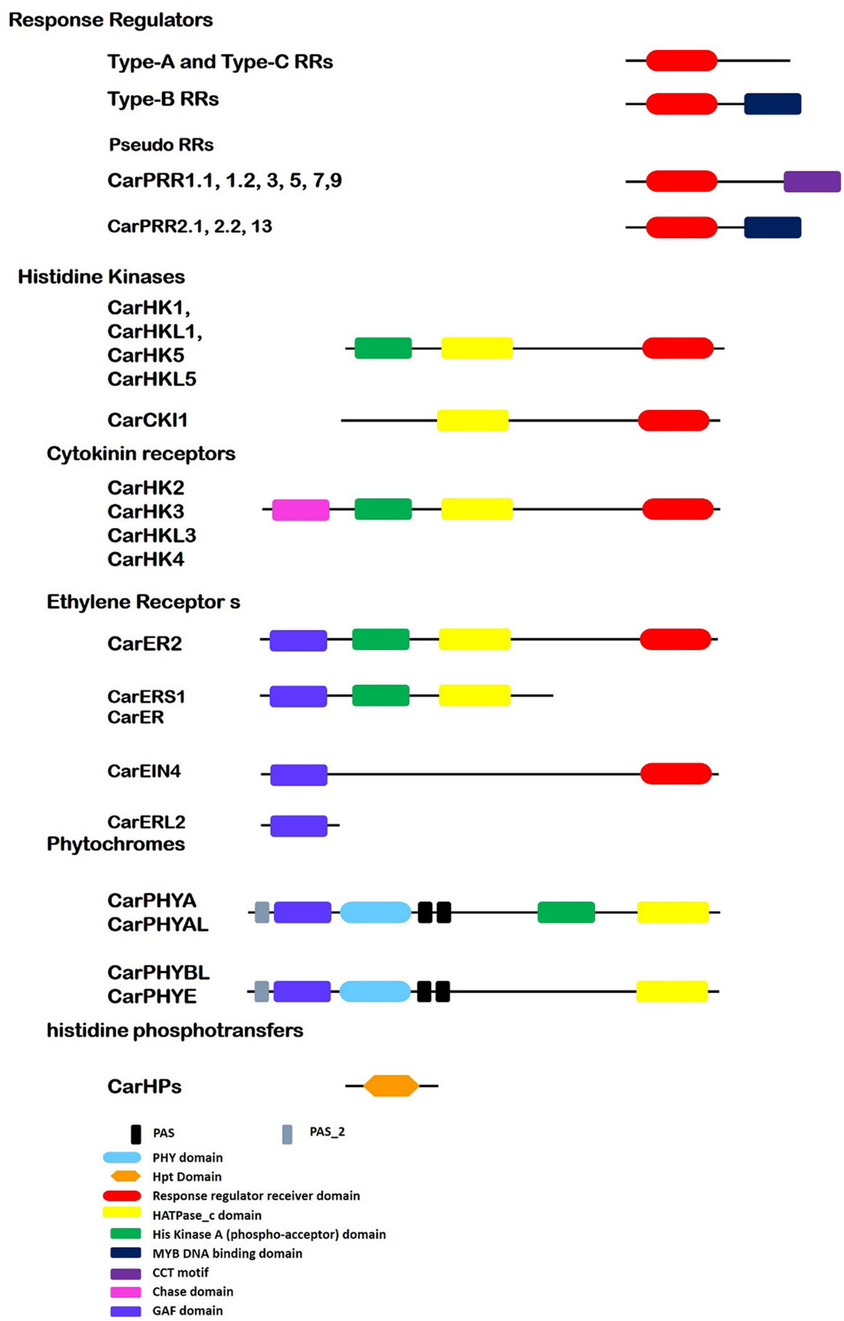


Fig. 1. Symbolic domain structures of chickpea two-component signaling elements.

Fig. 1).

In *Arabidopsis*, molecular and genetic studies revealed that there are five ethylene receptors (ERS1, RS22, EIN4, ETR1, and ETR2), which are involved in ethylene response [63]. Domain analysis showed that ethylene receptors contain a HisKa domain, a GAF protein-protein interactive domain and three N-terminal trans-membranes localized in plasma membrane. Likewise, five ethylene receptors were predicted in chickpea that displayed the same pattern of conserved domains as in *Arabidopsis*. CarER2 and CarEIN4 contain a C-terminal Rec domain that is absent in CarERS1 and CarER (Supp Fig. 1). ETR1 and ERS1 contain all five conserved motifs in *Arabidopsis*, whereas other members do not contain these conserved signatures. In chickpea, all members contain conserved Histidine (H) residue except CarER2 and CarERL2 where H is substituted by aspartic acid (D) and asparagine (N) respectively. It suggests that these are divergent HKs. All five motifs are conserved in CarER1 and CarEIN4, which also indicates that they are true HKs. CarERS1 contain all five conserved motifs except F motif in which phenylalanine (F) is replaced by isoleucine (I) (Supp Fig. 1).

Phytochrome family members, also known as photoreceptors, enable the plants to respond light stimuli and regulate the processes of growth and development [9]. There are five phytochrome members in *Arabidopsis* (PHYA, PHYB, PHYC, PHYD and PHYE). Domain analysis exhibited that phytochromes contain a GAF domain, an N-terminal PHY domain (involved in light absorption), two PAS domains and one Hiska domain involved in signal transduction [9]. Phytochromes are soluble proteins and contain same structure as sensor H protein kinases. They contain a C-terminal Hiska domain for signal transduction and an N-terminal sensor domain. They are referred to as divergent HKs as all five conserved motifs are absent in phytochrome family. They contain a new Ser/Thr kinase activity instead of HK activity [9,64]. Four phytochrome (PHYA, PHYAL, PHYBL and PHYE) members were predicted in chickpea (Table S1, Fig. 1). The CarPHYBL and CarPHYE do not contain

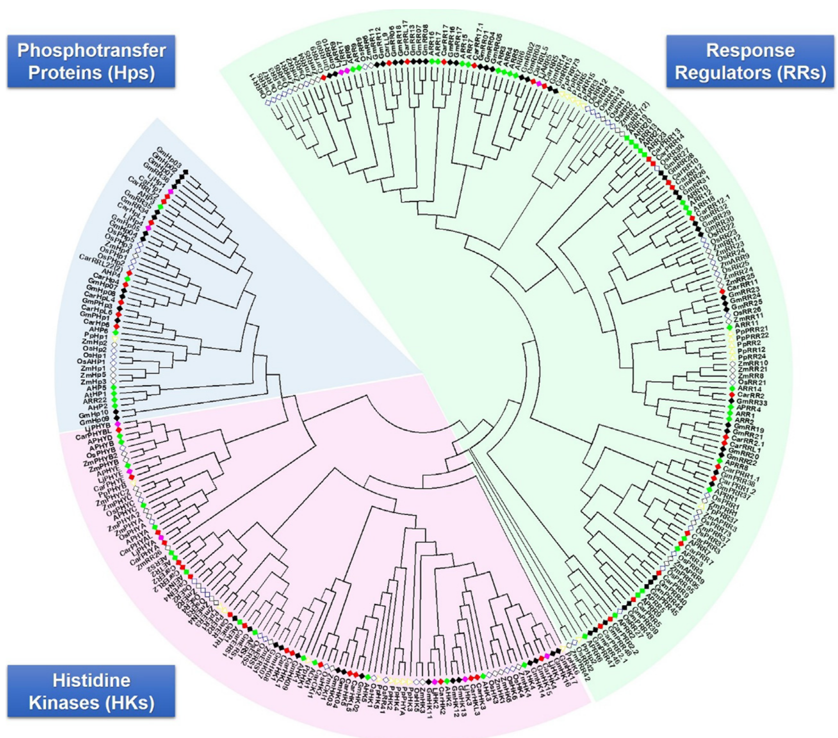


Fig. 2. The evolutionary history was inferred by using the Maximum Likelihood method based on the JTT matrix-based model. The tree with the highest log likelihood (-216493.7765) is shown. Initial tree(s) for the heuristic search were obtained automatically by applying Neighbor-Join and BioNJ algorithms to a matrix of pairwise distances estimated using a JTT model, and then selecting the topology with superior log likelihood value. The analysis involved 279 amino acid sequences. There were a total of 2659 positions in the final dataset. Evolutionary analyses were conducted in MEGA7 [50].

HisKa domain. They were declared as divergent HKs because all five conserved motifs were not found in their protein sequence (Supp Fig. 1).

3.1.2. HP family

Out of six *Arabidopsis* HP family members, five (AHP1-AHP5) are considered as true HPs and, one of them (AHP6) is declared as pseudo-Hp because it lacks a conserved H residue necessary to receive phosphate from the donor proteins [65,66]. All members contain a conserved phosphorylation motif (XHQXKGSSXS) except AHP6/APHP1. The H residue of phosphorylation motif is replaced by N in AHP6/APHP1. In *C. arietinum,* seven putative members of HP family were found, which contain a conserved Hpt domain (Table S2, Fig. 1). Five of these members (CarHp1, CarHpL1, CarHpL1.1, CarHp4 and CarHpL4) are classified as true HPs as they contain the conserved phosphorylation motif, whereas the other two members (CarHp6 and CarHpL6) are pseudo-Hps as they lack the conserved H residue in phosphorylation motif (which is replaced by N) (Supp Fig. 2).

3.1.3. The response regulators

Response regulators modulate the final response to the environmental stimuli. There are 32 members of RR family in *Arabidopsis* [23]. Alignment of Rec domain of RR family members in *Arabidopsis* showed that they contain conserved aspartic acid (D) and Lysine (K) residues (which is the major characteristic of response regulators). Basically, RR family members are divided into three subfamilies on the basis of conserved domains: type-A RRs contain a receiver domain with conserved D residue and long C-terminal extension, type-B RRs contain a conserved Rec domain and a DNA binding domain (Myb) and type-C RRs contain same domain structure as type-A RRs except the C-terminal extension. Furthermore, there is another class of RRs known as pseudo-response regulators (Divergent Response Regulators) that contains a C-terminal CCT motif and a conserved Rec domain in which conserved D residue is replaced by E 15. In current genome-wide analysis, 26 RR family members (7 type-A RRs, 7 type-B RRs, 2 type-C RRs and 10 pseudo-RRs) were found in *C. arietinum* (Table 1 and S3).

Seven type-A RRs were found in *C. arietinum* and this number is same as in L. *japonicas* and *P. patens*, whereas this number is relatively smaller than *O. sativa*, *Arabidopsis* and *G.* max (Table1 and S3). All the putative type-A RR family members showed a significant similarity to their counterparts in *Arabidopsis.* Domain analysis indicates that they contain a conserved Rec domain (Fig. 1) with highly conserved D-D-K motif, which is the final receiver of a phosphate group with a highly conserved D-residue at the N-terminal (Supp Fig. 3).

Type-B RR family members are nuclear localized proteins. They are different from type-A RR family members because they contain an additional DNA binding domain (Myb) [67,68]. It has been shown that type-B RRs function as transcription factors [67,69]. Seven type-B RR family members were found in *C. arietinum* and this number is relatively greater than *P. patens* and smaller as compared to *A.thaliana, O. sativa and G. max* (Table 1 and S3). Multiple sequence alignment of Rec domain indicates that they contain a conserved N-terminal D residue of D-D-K motif, which plays a key role in receiving the phosphate group to initiate the downstream processes (Supp Fig. 3).

There are two members (ARR22 and ARR24) in *Arabidopsis* that belong to type-C RR subgroup. They contain a conserved Rec domain like type-A RRs, but the C-terminal is very short. Phylogenetic analysis shows that they are not closely related to the type-A RRs [23,70]. It is also reported that they are not expressed in response to cytokinin [14,70]. Two members of type-C RR (CarRR22 and CarRRL22) family were found in *C. arietinum,* which show significant homology to their *Arabidopsis* counterparts and contain a conserved Rec domain with a short C-terminal.

3.1.4. Divergent response regulators

Divergent RRs or PRRs family is an additional group of RRs present in different plant species. There are nine members of divergent RRs in *Arabidopsis* (APRR1 to 9). They contain a complete Rec domain but D-D-K motif is not conserved [15]. Based on C-terminal extensions, they are subdivided into two groups: clock associated PRRs (APRR1/TOC1, APRR3, PRR5, PRR7, and PRR9) and type-B PRRs (APRR2, APRR4, and APRR6). Clock PPRs contain conserved basic amino acids (arginine (R) and lysine (K) in CCT motif. On the other hand, type-B PRRs contain a Myb domain at C-terminus instead of CCT motif. Ten PRRs are found in

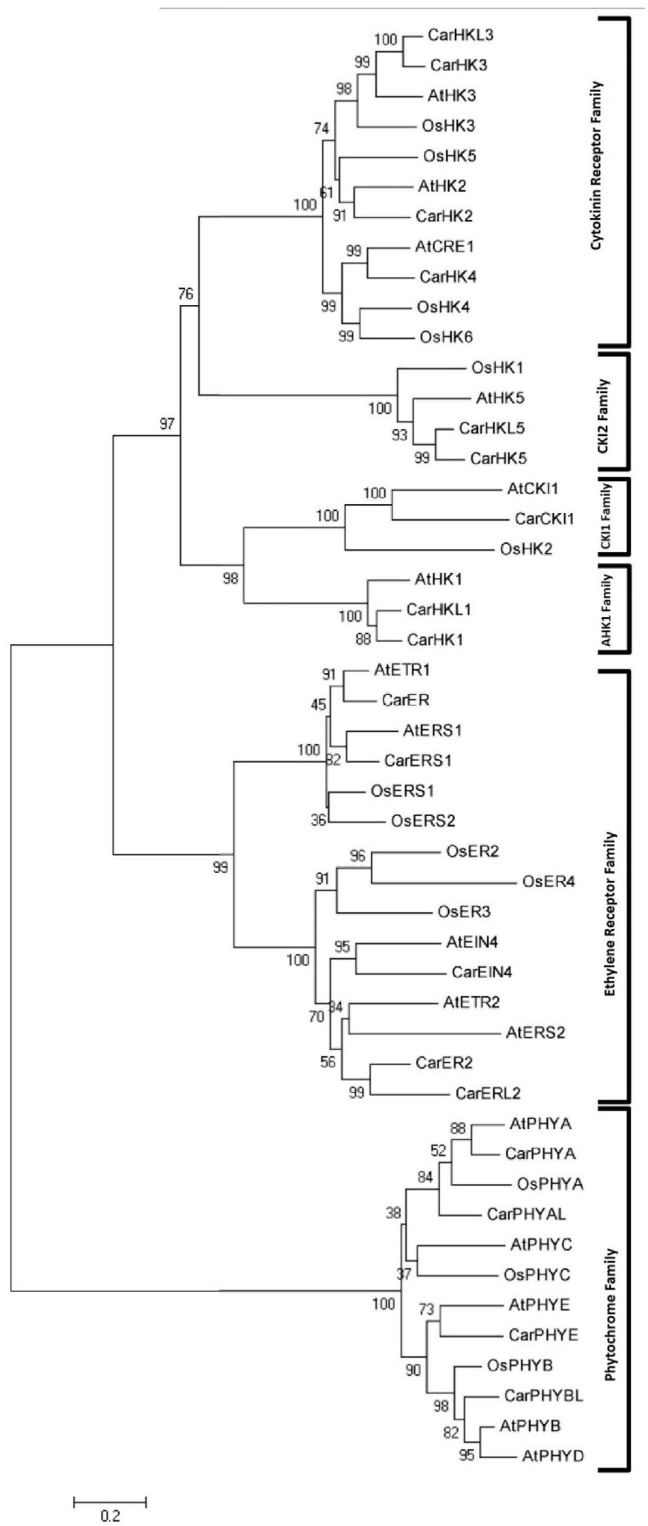


Fig. 3. Phylogenetic analysis of HK family members of *O. sativa, A. thaliana* and *C. arietinum.* The tree is derived by neighbor-joining method with bootstrap analysis (1000 replicates) from alignment of protein sequences. The tree is drawn to scale, with branch lengths measured in the number of substitutions per site. Evolutionary analyses were conducted in MEGA7 [50].

chickpea genome and this number is smaller than *G.* max and comparatively larger than *O. sativa*, *L. japonicas* and *Arabidopsis* (Table1 and S3. Fig. 1). CarPRR2.1, CarPRR2.2, Car PRR13 and CarPRR14 are considered as type-B PRRs (homologues of *Arabidopsis*PRR2, PRR4, and PRR6), while remaining members were classified as clock-associated PRRs. Multiple sequence alignment indicates that N-terminal D residue of D-D-K motif is substituted by glutamate (E) in CarrPRR2.1, CarrPRR2.2, CarPRR5, CarPRR7.1 and CarPRR9 and is conserved in CarPRR1.1, CarPRR1.1 and CarPRR7. Central residue of D-D-K motif is replaced by glutamic acid (E) in all members (Supp Fig. 3).

3.2. Phylogenetic analysis

Protein sequences of the TCS genes from *Arabidopsis*, *O. sativa*, *Z. mays, L. japonicas, G. max, P. patens* and *C. arietinum*, were used for sequence alignment and phylogenetic analysis (Fig. 2). These proteins were grouped into three major classes Histidine Kinases (HKs), Phosphotransfer Proteins (Hps) and Response Regulators (RRs). TCS members of *P. patens* (a moss model) were present on solitary branches (yellow triangles in Fig. 2). Likewise, TCS members belonging to mono and dicot species were also present on unilateral branches (empty/filled squares in Fig. 2). Similarly, TCS members belonging to leguminous and non-leguminous plants also occupied separate branches of phylogenetic tree. Such multi scale conservation of TCS protein sequences predicts functional conservation of these proteins as well. However, type-C RRs from *Arabisopsis*, *Z. mays, G. max* and *C. arietinum* shared branches, which diverged from HKs. It is in accordance to previous description of type-C RRs as the oldest RRs [77].According to phylogenetic tree, HKs were divided into six subgroups (cytokinin receptor, AHK1-like, CKI1-like, CKI2- like, ethylene receptor and phytochrome subfamily) (Fig. 3). All the HKs share a clade with dicot plant *Arabidopsis,* which indicates emergence of HKs after the divergence of mono- and dicotyledon plant species. There are three cytokinin receptors (AHK2, AHK3, and AHK4) in *Arabidopsis* [1,2], while 4 (CarHK2, CarHK3, CarHKL3 and CarHK4) in *C. arietinum.* Phylogenetic analysis indicated that CarHK3 and CarHKL3 are co-orthologue of AtHK3, while CarHK2 is a direct orthologue of AtHK2. Functional characterization of cytokinin receptors in *Arabidopsis* showed that they are involved in a number of cytokinin regulated processes like stress responses, leaf senescence, cell division, seed size and vascular differentiation [34]. Based on domain structure and phylogenetic analysis, it could be speculated that the putative cytokinin receptors in *C. arietinum* perform the same functions as their counterparts in *Arabidopsis.* However, presence of an extra cytokinin receptor in *C. arietinum* hints a potential functional redundancy or diversity.

AHK1 is a transmembrane localized protein, involved in osmosensing, abundantly expressed in roots of *Arabidopsis* [71]. It plays a positive regulatory in salt stress conditions [34]. There are two co-orthologues of AHK1 in *C. arietinum* (CarHK1 and CarHKL1). In CKI2 group, phylogenetic indicated that CarHK5 and CarHKL5 are co-orthologue of CKI2 (Fig. 3). The ethylene receptors have been divided into two groups. Group I contains AtETR1 and AtERS1 along with CarER and CarERS1 as their direct homologues, respectively. In group II, ETR2, ERS2 and EIN4 like proteins are present on different clades for monocots (*O. sativa*) and dicots (*A. thaliana and C. arietinum*). CarER2 and CarERL2 are paralogues in chickpea and show an orthologous relation with AtER2 of *Arabidopsis*. While, CarEIN4 is the direct orthologue of EIN4. The members of phytochrome family (CarPHYA, CarPHYAL, CarPHYBL and CarPHYE) show more close relationship with monocotyledon plant *O. sativa* (Fig. 3). It indicates that PHY-like proteins were formed before divergence of mono and dicotyledons.

Phytochromes also known as photoreceptors are involved in response to light, growth and development [71-73]. Five photoreceptors are present in *Arabidopsis,* which contain an N-terminal PHY domain, C-terminal GAF domain, two PAS folds, and one Hiska domain involved in signal transduction. Four photoreceptors (CarPHYA, CarPHYAL,

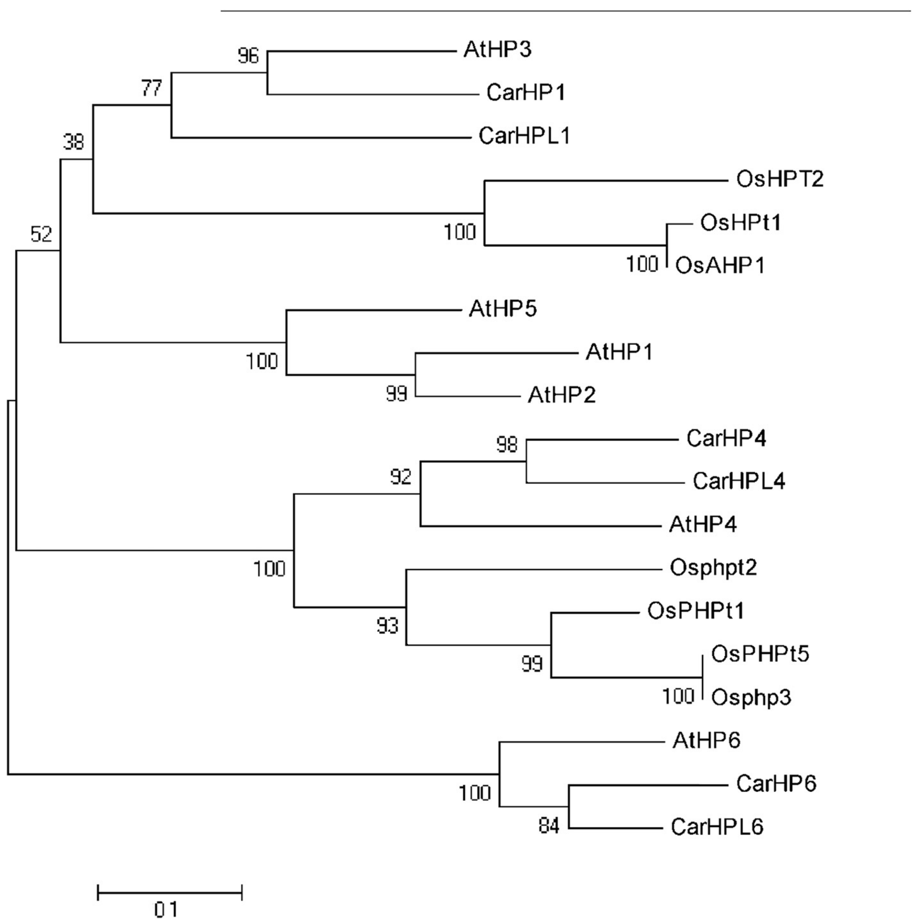


Fig. 4. Phylogenetic analysis of HP family members of *O. sativa, A. thaliana* and *C. arietinum*. The tree is derived by neighbor-joining method with bootstrap analysis (1000 replicates) from alignment of protein sequences. The tree is drawn to scale, with branch lengths measured in the number of substitutions per site. Evolutionary analyses were conducted in MEGA7 [50].

CarPHYBL and CarPHYE) were found in *C. arietinum,* which contain the same domain pattern as their counterparts in *Arabidopsis.* However, Car-phytochromes are homologues to only three At-phytochromes. Moreover, these members lack the conserved motifs necessary for normal functionality of HKs, so they were considered as divergent HKs. Phylogenetic analysis revealed that photoreceptors of *C. arietinum* are othologous and co-ortholgous of photoreceptors in *Arabidopsis* and *O. sativa.* Functional analysis of photoreceptors in *Arabidopsis* indicated that they contain ser/thr kinase activity instead of HK activity. Due to difference in number and homology, it is speculated that Car-phytochromes may perform diverse roles as compared to At-homologues.

There are two subgroups of HP family: true HPs and Pseudo-HPs. CarHP1, CarHPL1, CarHP4 and CarHL4 shows a closer relationship to the true Hps present in *Arabidopsis* and *O. sativa*. CarHP6 and CarHPL6 are co-orthologues of AtHP6, which predicts that both are potentially pseudo (divergent) Hps like AtHP6. CarHP6 and CarHPL6 also lack the conserved motif necessary for normal activity of HPs. CarHP1 and CarHPL1 show a close relationship to AtHP3, (a positive regulator of cytokinin signaling), which indicates potential involvement of both proteins in cytokinin signaling (Fig. 4). Seven HPs were found in *C. arietinum* in which five were classified as true HPs and two (CarHP6 and CarHPL6) were placed in the pseudo HP group.

For phylogenetic analysis of RRs, protein sequences of RRs in *Arabidopsis*, *O. sativa* and *C. arietinum* were used. RRs are divided into type-A RRs, type-B RRs, type-C RRs and PRRs (divergent RRs). In the present study, type-B RRs were further divided into type-B1 RRs, type-B2 RRs and type-B3 RRs (Fig. 5). PRRs are also classified as type-B PRRs and clock-associated PRRs. All the type-A RRs members show a close relation to their counterparts in *Arabidopsis* and *O. sativa*, which provides us a confirmation that they are true type-A RRs. The RR family is third partner of TCS system, type-A RRs, type-B RRs and type-C RRs, are involved in regulation of final response to the stimuli. Besides, there is another fourth group of RRs which is not considered as true-RR because D-D-K motif is not conserved among them and they referred as PRRs. Total 26 RRs (7 type-A, 7 type-B, 2 type-C and 10 PRRs) were found in *C. arietinum*. This number is smaller than the RRs present in *O. sativa, Arabidopsis* and *G.* max but greater than the RRs present in L. *japonicas* and *P. patens.* Type-A RRs are considered as new members of the RR family because they were observed first time in land plants and there is no evidence of their presence in unicellular algae. So, it could be suggested that they are specifically present in plants and perform some novel functions. The overall structure of the type-A RRs contain a Rec domain with a conserved D-D-K motif, necessary for receiving the phosphate group, and a long C-terminal extension. They are mostly activated by cytokinin and their cytokinin induction is partially dependent on the type-B RRs [67,68,74,75]. Phylogenetic analysis revealed that type-A RRs of *C. arietinum* are direct and co-orthologue of type-A RRs present in *Arabidopsis* and *O. sativa.*

All the members of type-B RRs showed a close relationship with the type-B1 RRs and no member was seen in close relationship with type-B2 (ARR13, ARR21, and ARR23 in *Arabidopsis*) and type-B3 members. Type-B RRs are oldest members of RR family because they are widely studied in unicellular algae as compared to type-A RRs [76].

Type-C RRs (CarRR22 and CarRR22L) showed a close relation with the *Arabidopsis* (ARR22) and rice counterparts (OsRR41 and OsRR42), which is a confirmation of true type-C RRs. Two members of type-C RR (CarRRL22 and CarL22) were also found in *C. arietinum,* which contain the same domain pattern like type-A RRs except the C-terminal extension which is missing in them. They are co-orthologue of member of type-C RR (ARR22) in *Arabidopsis*.

Members of type-B PRRs (CarPPR2.1, CarPRR2.2, CarPRR13 and CarPRR14) were seen to be in a close relation with their *Arabidopsis*.

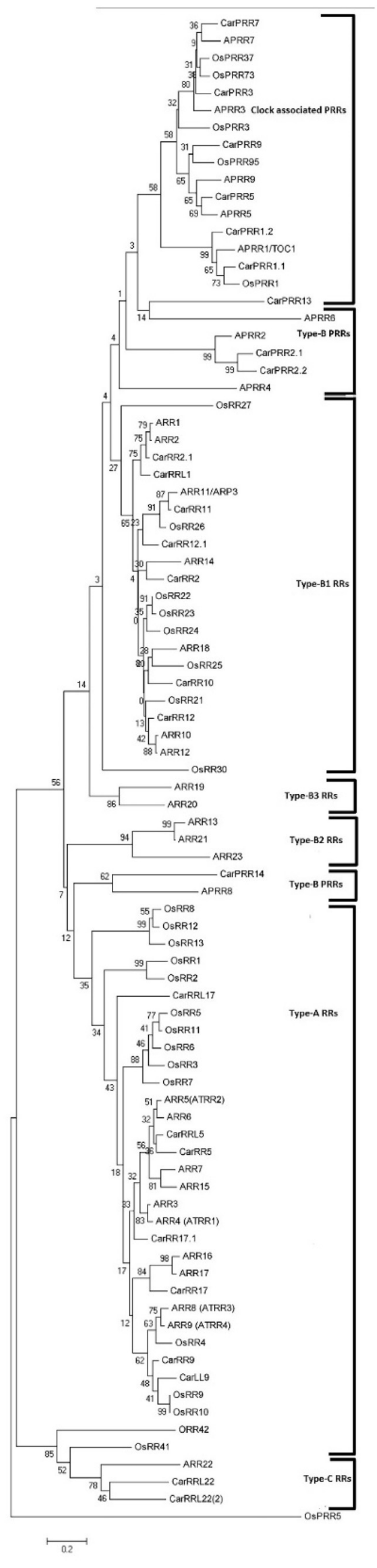


Fig. 5. Phylogenetic analysis of RRs and PRRs family members of *O. sativa, A. thaliana* and *C. arietinum*. The tree is derived by neighbor-joining method with bootstrap analysis (1000 replicates) from alignment of protein sequences. The tree is drawn to scale, with branch lengths measured in the number of substitutions per site. Evolutionary analyses were conducted in MEGA7 [50].

CarPRR13 and CarPRR14 were also found in a close relation with the reported type-B PRRs (APRR6 and APRR8) in *Arabidopsis*. CarPRR2.1 and CarPRR2.2 showed a close relation with APRR2, which confirms that CarPPR2.1, CarPRR2.2, CarPRR13 and CarPRR14 belong to type-B PRRs subfamily (Fig. 5).

3.3. Genomic distribution, evolutionary analysis and promoter analysis of the TCS members in C. arietinum

Chromosomal location of the TCS genes was determined from their genomic sequences, which were taken from NCBI and CTDB. Moreover, all the TCS genes were mapped on the eight chromosomes of *C. arietinum* except *CarER* and *CarPRR13* because their locus information was not available. Distribution of genes is uneven as maximum numbers of genes (10 genes) is present on chromosome number 4 and only one gene is present at chromosome number 2 (CarHK2) (Fig. 6).

Gene duplication plays a critical role in evolution of gene family as it provides a raw material for generation of a new gene. Number of genes in plants increases due to tandem or segmental duplication of genes [77]. Current study investigates the tandem and segmental duplication of the TCS genes. Multiple gene pairs exhibited segmental duplication and only five gene pairs were tandemly duplicated. Therefore, it can be implied that expansion of the TCS genes in *C. arietinum* is mainly due to segmental duplication (Table 2). This model of TCS gene expansion also exists in other plants like *Arabidopsis*, chinese cabbage, cucumber, watermelon and soybean [9,24,59,83, 84].

The gene divergence mechanism was calculated by finding the synonymous (Ks) and nonsynonymous (Ka) substitution rate. To explain the divergence time and evolutionary pattern of paralogous in *C. arietinum*, Ks and Ka modes for segmental duplication were calculated (Table S5). According to divergence time, the first duplication event occurred 256.7 mya in which a segment containing a *CarRR12* gene duplicated from chromosome 5 to 7 or vice versa. Latest duplication event took place 38.90 million years ago in which *CarHK5* duplicated through tandem duplication and produced a new gene *CarHKL5*.

To get an insight of potential transcriptional regulation of these genes, region upstream to transcriptional start site (1000-1200 bp) was searched for *cis*-regulatory elements. Number of abiotic stress-related (drought, light, high salinity, cold and extreme temperatures) and hormone-related (GA, Auxin, Meja, Ethylene, ABA, Gibberellins and SA) *cis* regulatory elements were predicted in promoter sequences of TCS genes (Table S4).

3.4. Expression analysis of CarTCS genes

For better understanding of the physiological role of TCS genes, the tissue-specific expression of TCS genes was determined in different tissues (shoot, root, mature leaf, flower bud and young pod) by analyzing the RNA-seq data available at chickpea transcriptome database (CTDB) [55]. It was observed that only 37 TCS genes were found to be expressed across five tissues in normal conditions (as shown in Fig. 7a, which indicates only presence or absence of gene expression). *CarHK2, CarHK3, CarHKL3* and *CarHK4* expressed globally in all tissues while the expression of *CarHK1* was detected in roots and shoots, while *CarHK5* in shoots and pods only. Among ethylene receptors, *CarERS1* and *CarER2* expressed in all tissues while *CarEIN4* was expressed in all tissues except flower buds. Similarly, *CarHP1* was the only phosphotransfer protein that was globally expressed. While the expression of *CarHPL1.1* was detected in flower buds/shoots and *CarHPL1* was

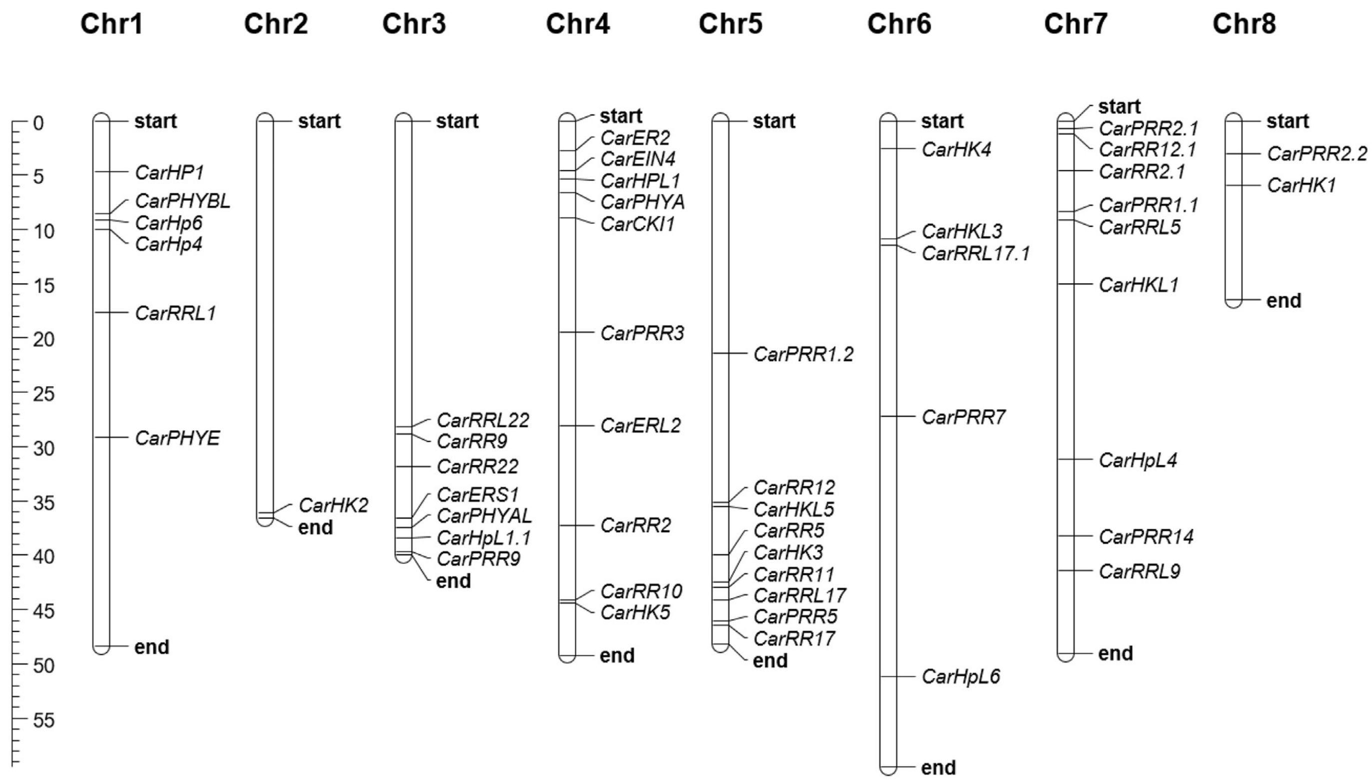


Fig. 6. Chromosomal mapping of TCS genes on eight chromosomes of *C. arietinum.* The scale at the left side of the chromosomal bar indicates the position on the chromosome (mega base pairs; Mb).

Table 2
Comparison of duplication of TCS elements in *Arabidopsis* and *C. arietinum.*

Gene family	Genes involved in Segmental Duplication	Percentage (%)	Genes involved in tandem duplication	Percentage(%)
CarHKs/AtHKs	10/2	55.55/12.5	4/0	22.22/0
CarHPs/AtHPs	6/2	85.71/33.3	0/0	0/0
CarARRs/AtARRs	4/10	57.14/100	0/0	0/0
CarBRRs/AtBRRs	4/6	50/50	0/0	0/0
CarCRRs/AtCRRs	0/0	0/0	2/0	100/0
CarPRRs/AtPRRs	8/0	80/0	0/0	0/0

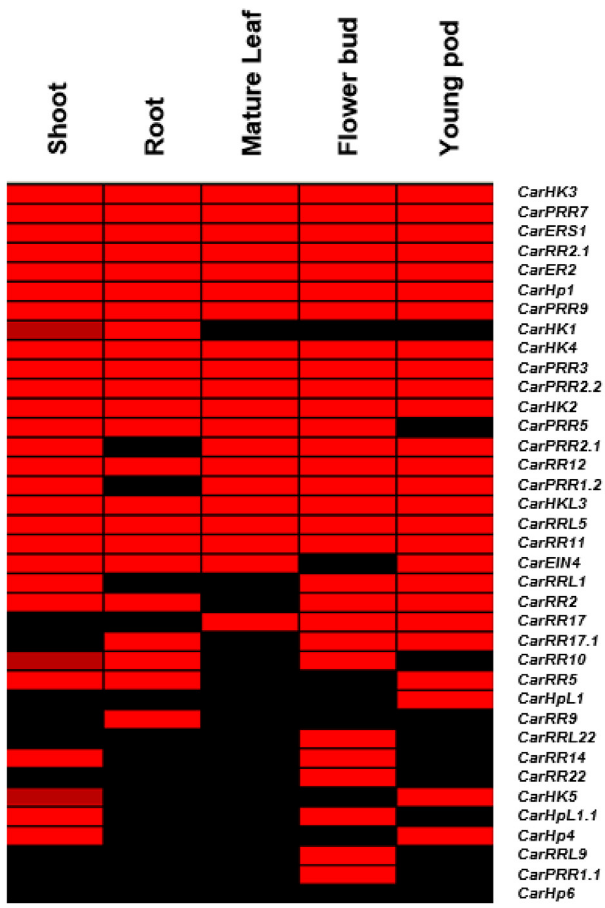
Car representing *C. arietinum* while At represents *A. thaliana.*
CarHKs (HKs in *C. arietinum*)
CarARRs(Type-A RRs in C. arietinum)
CarBRRs (*Type-A RRs in C. arietinum*)
CarCRRs (Type-C RRs in C. arietinum)
CarPRRs (PRRs in C. arietinum)

expressed only in young pods. Response regulators exhibited diverse expression in different tissues. *CarRR9* specifically expressed only in roots while *CarPRR1.1, CarRRL9, CarRR22* and *CarRRL22* expressed only in flower buds. Type-C RRs show a similar expression pattern and are highly expressed in flowers as in *Arabidopsis* [14,70].

From CTDB, expression data was also analyzed for abiotic stress (drought, salinity and cold stress) response of TCS genes (Fig. 7b). The expression of different genes was modulated in response to stress factors such as *CarRR12,* which showed a lower expression across shoots in response to salt stress and a higher expression in response to drought stress. Several genes such as *PRRs,* were down regulated across the roots in response to cold and drought stress. Expression of *CarHK1* is also decreased in shoots in response to cold and salt stress (Fig. 7b). Expression of type-B RRs, which act as transcription factors, was not significantly changed in response to different abiotic stresses.

To confirm the expression data under stress conditions in our lab, relative Real Time-qPCR was performed for selected genes (Table S4) in response to drought, heat, osmotic and salt stress. The results of qPCR (Fig. 8) were mostly correlated with RNA-seq expression data (Fig. 7b). However, some differences were also found (expression of CarHp1). Although, we apply similar growth conditions, such differences could be related to plant age, genotypic differences or even sampling time in a day. It was observed that the transcript abundance of *CarHK, CarHK1, CarHKL1, CarPHYA, CarERS1 CarHP1, CarRR5, CarRR12, CarRR12(2), CarPR17, CarPRR1* and *CarPRR2* is significantly modulated by different abiotic stress conditions (Fig. 8). It is worth to highlight that expression of *CarHKL1, CarPHYA, CarRR2* and *CarRR12* is most significantly upregulated in response to drought stress (Fig. 8). Similarly, the expression of *CarHK1* and *CarRR17* is upregulated and *CarRR12.1* is downregulated in response to heat stress. In response to mannitol stress, *CarHK1, CarRR12* and *CarRR17* expression is upregulated, while expression of *CarHP1, CarRR5* and *CarRR12.1* is downregulated. Likewise, expression of *CarHK1, CarPHYA, CarERS1,* and *CarRR17* is upregulated, and *CarHP1* is downregulated in response to salt stress (Fig. 8).

In *A. thaliana*, *AHK1* is involved in abscisic acid (ABA) signaling and a)



b)

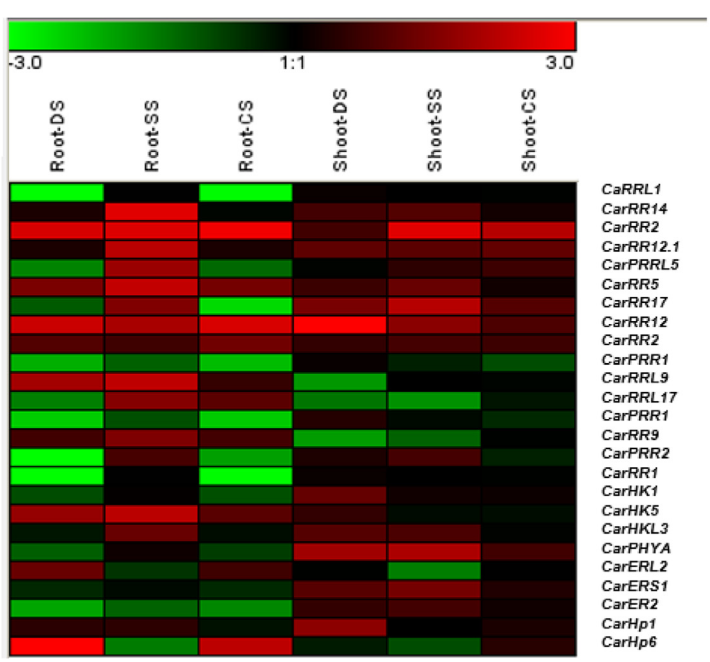


Fig. 7. Gene expression of TCS genes. a) Representation of tissue specific expression of TCS genes in normal conditions. Tissue samples were collected from shoots and roots of 15 day old seedlings. The young pods, flower buds and mature leaves were gathered from field grown plants [80]. Red color represents the presence of a gene expression while black color represents an absence of that gene. b) Heat map representation of tissue specific expression of TCS genes in cold (CS), salt (SS) and drought (DS) stress conditions. The seeds of chickpea were grown in a culture room at 22 ± 1 °C. For salinity and drought stress application, seedlings (at the age of 10 days) were transferred in 150 mM NaCl solution and on folds of tissue paper, respectively. For cold stress, the seedlings were kept in water at 4 ± 1 °C. The samples were collected 5 h after treatments [85-87]. Expression data is presented as fold-change by comparing with the corresponding non-treated samples. Red color represents an up-regulation, green color for down regulation and black color represents no change in gene expression. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

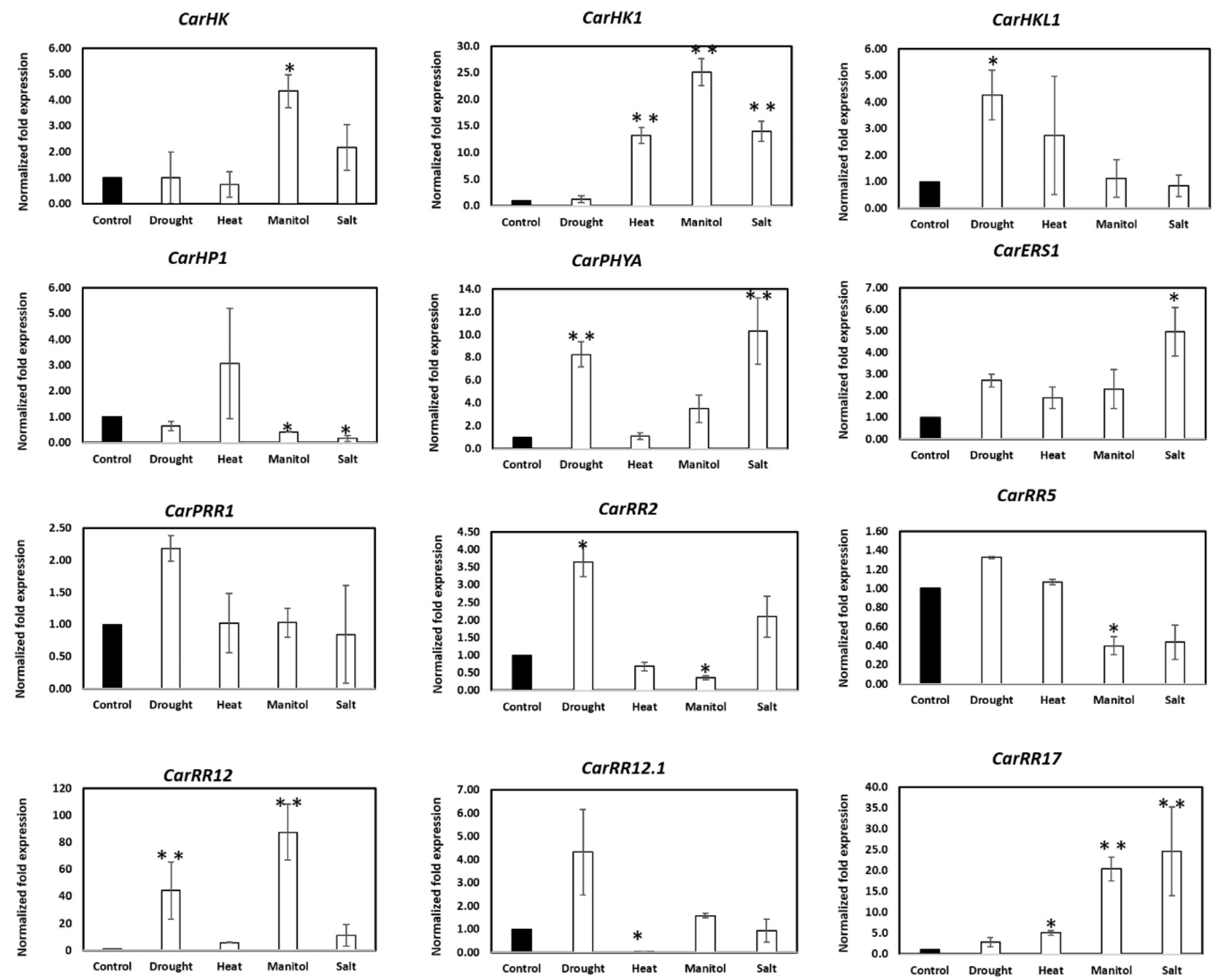


Fig. 8. Relative qRT-PCR assay of TCS genes under drought, heat, osmotic and salt stress. The default expression value for each gene was one in non-treated plants. Error bars indicate the standard deviation from mean (three replicates). Asterisks on the top of bars indicate statistically significant differences between treated and non-treated plants with a *P*-value ≤.05 (*) or 0.01 (**).

as an osmosensor [72,78]. In current study, the expression of *CarAHK1* highly correlates with its potential role in drought stress and osmosensing. It was recently reported that *ahk* mutants exhibit particular phenotypes associated with AHK2 and AHK3 in *Arabidopsis.* Furthermore, these phenotypes were also linked with cytokinin signaling for accumulation of proline and induction of *NCED3* and *P5CS1* gene expression under low water potential conditions [82]. Interestingly, *CarHP1* expression was reduced in response to osmotic and salt stress (Fig. 8). Meanwhile, characterization of *phyA*, *phyB* and *phyAB* mutants displayed enhanced tolerance against salt stress as compared with wild-type tobacco plants [83]. PHYA expression in current study predicts contrary role for these phytochromes, which may be associated with species specific genotypic differences. However, detailed functional characterization is required to define exact role of these kinases.

In the same way, type-B RRs perform a redundant but essential function in cytokinin signaling during plant life cycle [35]. Moreover, type-B RRs are involved in transcriptional regulation of type-A RRs [71, 79,80]. Expression analysis demonstrated that the expression of type-B RRs (*CarRR2* and *CarRR12*) is significantly increased in response to drought and osmotic stress (Fig. 8). It indicates that type-B RRs are involved in drought stress response like their *Arabidopsis* homologues [81]. Moreover, CarRR17 (Type-A RR) was significantly upregulated in response to heat, mannitol and salt stress (Fig. 8). Recently, it was reported that Type-A RRs may act as regulators of the HK1 phosphorelay pathway [85, 87]. Thus, we propose that the two-component system is structurally and functionally conserved between Chickpea and *Arabidopsis* [57,69]. However, there are some variations that may be studied in detail to define the role of TCS genes in plant stress response.

4. Conclusion

This study reported 51 putative members of TCS protein family including 18HKs, 26 RRs (7 type-A, 7 type-B, 2 type C and 10 pseudo) and 7 HPs (5 true and 2pseudo) in *C. arietinum.* These TCS proteins exhibited a close phylogenetic relationship with TCS of other legumes. Since, these genes exhibit highly conserved characteristics in heterologous systems; it is predictable that *TCS* genes perform comparable functions in diverse taxonomic groups. According to RNA seq-data, most of TCS genes were expressed in flower buds and respond to environmental stresses. Meanwhile, promoter analyses and qRT-PCR results also indicated that TCSs could respond to major environmental stresses. Current foundation data is expected to facilitate future studies for understanding functional importance of *TCS* genes.

Author contributions

For research articles with several authors, a short paragraph specifying their individual contributions must be provided. The following statements should be used "Conceptualization, F.A. and M.A.A.; Methodology, B.A. and M.A.A.; Software, F.A. and M.A.N.; Validation, F.A., M.A.A., H. N. and G. C.; Formal Analysis, B.A., F.A., A.A., and U.I.; Investigation, B.A. and RMA; Resources, M.A.A. and G.C.; Data Curation, B.A.; Writing-Original Draft Preparation, F.A., M.N.C. and M.A.A.; Writing-Review & Editing, M.A.N., G.C. and R.B.; Visualization, M.A.A.; Supervision, M.A.A. and G.C.; Project Administration, F.A. and G.C.; Funding Acquisition, F.A. and G.C.",

Funding

This research was funded by International Foundation for Science (IFS) (grant number no. C/5684-1). This work was also supported by the Cooperative Research Program for Agriculture Science and Technology Development (Project No. PJ014344) Rural Development Administration, Republic of Korea.

Declaration of Competing Interest

The authors declare no conflict of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ygeno.2019.08.006.

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